SIT723 Research Techniques & Applications

Task 2.3D Initial Literature Review

Literature Review

Large language models (LLMs) have demonstrated remarkable ability in natural language processing tasks such as reasoning, question answering, and biomedical applications. However, a growing body of evidence suggests that high benchmark accuracy does not necessarily indicate stable semantic understanding: when the same meaning is expressed through slightly different prompts, paraphrases, or syntactic structures, models can produce different outputs, which calls into question whether current evaluation methods measure reliable interpretation. This concern is particularly acute in medicine, where unstable outputs directly affect the reliability of clinical language technologies. The review below synthesises the main literature themes relevant to this project and shows how they converge on the proposed research direction: a UMLS-grounded semantic entropy framework for evaluating semantic stability in clinical NLP.

A. Prompt Sensitivity and Semantic Consistency

A first strand of the literature establishes that LLM performance is highly sensitive to prompt wording and formatting. Sclar et al. [5] showed that purely cosmetic changes to prompt formatting, separators, spacing, and option labels, none of which alter the task, can produce accuracy spreads of up to roughly 76 percentage points across templates on multiple-choice benchmarks with open-weight LLMs such as Llama-2, measured directly as the accuracy range across format variants. Zhuo et al. [6] complement this with ProSA and the PromptSensiScore (PSS), a reusable instance-level metric that quantifies how model outputs shift under semantically equivalent but differently worded prompts; evaluated across arithmetic reasoning (GSM8K), commonsense reasoning (CommonsenseQA), and challenge reasoning (ARC-Challenge), they found that prompt sensitivity varies substantially across tasks and models rather than being a fixed model-wide property, and that incorporating few-shot demonstrations alleviates but does not eliminate the effect. Relative to [5], whose contribution is the scale of the observed formatting sensitivity, [6] adds a reproducible measurement framework operating at the instance level; a shared limitation is that both studies use general-domain benchmarks and neither addresses the clinical setting targeted by the present project.

Semantic consistency studies sharpen this picture by focusing on paraphrase rather than formatting. Raj et al. [8] evaluated open-source LLMs on paraphrased questions and reported that semantic inconsistency, measured as the proportion of disagreeing answers across meaning-equivalent prompts, is substantially higher than the task error rate, separating one-shot correctness from stable interpretation. Wang et al. [7] demonstrated with self-consistency decoding that sampling multiple chain-of-thought paths and taking the majority answer raised PaLM-540B from 56.5% to 74.4% accuracy on GSM8K, direct evidence that inter-sample agreement correlates with correctness. Wei et al. [35] provided the underlying chain-of-thought result, reporting that step-by-step prompting lifted PaLM-540B on GSM8K from 17.9% to 56.9% accuracy, underscoring how strongly output behaviour depends on prompt structure. Novikova et al. [24] then surveyed factual, hypothetical, compositional, and moral consistency, and noted that no single benchmark covers all facets, a methodological weakness that is visible in every study from Sclar et al. through Wei et al., as no common protocol for stability is used.

Read together, these studies show that prompt sensitivity and semantic inconsistency are not isolated failure modes but a systemic evaluation problem. Formatting and prompt design can swing measured performance dramatically even without any meaning change [5], paraphrase can flip answers when the underlying question is semantically equivalent [8], and the broader consistency landscape remains fragmented across incompatible criteria [24]. The common thread is that current practice rewards the final answer while leaving the stability of the underlying interpretation largely untested, which is the evaluation gap this project targets.

B. Adversarial Robustness and Meaning-Preserving Perturbations

Classical adversarial NLP research supplies the methodological toolkit for probing instability under minimal, meaning-preserving input changes. Jin et al. [25] introduced TextFooler, a contextual synonym-substitution attack that preserves sentence semantics and grammaticality; on IMDB sentiment classification it reduced BERT accuracy from approximately 92.2% to 6.6% while perturbing fewer than 20% of words per example, evaluated as post-attack accuracy and perturbation rate. Ebrahimi et al. [26] with HotFlip used gradient-guided character flips to drop classifier accuracy on AG News with only a handful of edits per document, showing that even character-level perturbations suffice to change predictions. Of the two, [25] is the more direct precedent for the present study because it explicitly enforces semantic preservation, whereas [26] emphasises surface-level fragility; together they justify restricting perturbations to meaning-preserving transformations so that any output drift can be attributed to the model rather than to the input.

The LLM-era literature confirms and extends these findings but exposes new methodological issues. Goyal et al. [9] surveyed adversarial defences in NLP and argued that attack success must be separated from semantic preservation, because many headline attack-success numbers partly reflect meaning change rather than true model fragility, a conceptual strength that this project directly inherits. Alahmari et al. [10] evaluated GPT-3.5, GPT-4, and Llama-2 variants under several perturbation types and reported that larger models are only modestly more robust, with accuracy drops still in the low-to-mid double digits under synonym and paraphrase attacks, measured as pre- versus post-perturbation accuracy. Xu et al. [11] introduced PromptAttack and showed that an LLM can generate adversarial prompts capable of misleading itself and peer models, indicating that vulnerability is partly endogenous to generation. Şahin [32] compared text augmentation techniques, back-translation, synonym replacement, and noise insertion, and found that no single method dominates across low-resource NLP tasks, with effectiveness measured by downstream F1; the methodological implication is that perturbation choices must be justified on semantic grounds rather than by convention.

Within the biomedical domain, Moradi and Samwald [36] showed that adversarial training on biomedical language models (BioBERT and BlueBERT-style models) improved micro-F1 by several points on benchmarks such as BC5CDR and i2b2 while also improving robustness under synonym-level perturbation, evaluated as pre- versus post-attack F1. Compared with the general-domain attacks of Jin et al. and Ebrahimi et al. [25, 26], Moradi and Samwald [36] is the only study in this cluster conducted on biomedical text and therefore provides the closest precedent for the present project; its weakness, which it shares with the recent LLM adversarial work discussed above, is that stability is measured only through task accuracy rather than through a meaning-level uncertainty metric, a gap taken up in Theme C.

C. Uncertainty Estimation and Semantic Entropy

Traditional uncertainty work focuses on output-probability calibration. Guo et al. [30] showed that modern deep networks are systematically overconfident, reporting an expected calibration error (ECE) of 16.53% for ResNet-110 on CIFAR-100 against the ideal of ~0%, and demonstrated that temperature scaling reduces ECE to 1.26% without changing accuracy; this establishes that probabilities require post-hoc correction but does not imply that they track meaning. Kadavath et al. [23] extended calibration analysis to LLMs with the P(IK) self-evaluation metric and found that aggregate calibration improves with scale while instance-level self-assessment remains unreliable, which is particularly concerning for high-stakes applications. Geifman and El-Yaniv [31] approached reliability via selective classification and reported coverage–risk trade-offs in which a fixed error target is maintained by abstaining on 10–30% of inputs, depending on the dataset. The shared weakness of this calibration-and-abstention line is that probability-based signals do not measure whether a model's interpretation is invariant across semantically equivalent inputs: temperature-scaled softmax values [30] are tuned on correctness rather than meaning, instance-level P(IK) self-assessment is unreliable even in very large models [23], and selective prediction [31] controls abstention rather than stability.

Semantic entropy addresses that gap directly. Kuhn et al. [1] defined semantic entropy over clusters of semantically equivalent generations, identified using a bidirectional natural-language-inference (NLI) model, and reported AUROC improvements over length-normalised token-entropy baselines for uncertainty-based selective answering on TriviaQA and CoQA. Farquhar et al. [2] scaled this framework in Nature across multiple LLMs and QA datasets, showing that semantic entropy outperforms both token-level log-likelihood and lexical-similarity baselines at detecting hallucinated answers with AUROC gains on the order of 5–10 points depending on the model; its strength is the principled equivalence-based formulation, and its remaining weakness is that equivalence is operationalised through a general-domain NLI model rather than an ontology. Savage et al. [20] applied consistency-based uncertainty proxies to GPT-4 on medical QA and reported improved discrimination (AUROC) and calibration (Brier score) for diagnostic and therapeutic reasoning questions, which transfers the stability argument into the clinical domain. Manakul et al. [40] proposed SelfCheckGPT as a zero-resource alternative that scores a response against multiple self-samples and reported that its agreement scores correlate with human-annotated hallucination labels on the WikiBio dataset; it is attractive because it requires no external equivalence model, but compared with the semantic-entropy formulation of Kuhn et al. and Farquhar et al. [1, 2] it lacks a formal semantic-equivalence criterion and therefore functions as a natural baseline rather than a competitor.

Taken together, the meaning-level uncertainty line of work, semantic entropy [1] and its large-scale validation for hallucination detection [2], consistency-based uncertainty proxies in medical QA [20], and the zero-resource consistency check of SelfCheckGPT [40], moves the unit of uncertainty from tokens to meanings. The earlier calibration and selective-prediction work discussed above [30, 31] still supplies the reliability backdrop but does not address meaning preservation. The main outstanding weakness in this literature is that semantic equivalence is typically approximated by an NLI model or by self-similarity. Grounding the equivalence relation in a biomedical ontology, as this project proposes, converts an approximate criterion into a formal one and is the specific contribution anticipated here.

D. Biomedical Language Models, UMLS, and Concept-Level Evaluation

Domain-specific models justify treating clinical NLP as a distinct evaluation setting. Lee et al. [14] showed that BioBERT, pretrained on PubMed abstracts and PMC full-text articles, raised F1 by roughly 0.62 points on biomedical NER and by 12.24 MRR points on biomedical QA compared with prior state-of-the-art models, establishing the empirical value of biomedical pretraining. Gu et al. [12] extended this line with PubMedBERT, trained from scratch on biomedical text, and reported new state-of-the-art results across the BLURB benchmark, with micro-F1 gains over mixed-domain baselines on tasks such as BC5-chem, BC5-disease, and ChemProt; their core methodological claim that mixed-domain pretraining is suboptimal for biomedical language is directly relevant because this project relies on concept-level rather than token-level semantics. Tinn et al. [13] studied fine-tuning stability of PubMedBERT and showed that biomedical models differ not only in accuracy but also in variance across random seeds and hyperparameters, a finding conceptually adjacent to stability under semantic perturbation. Labrak et al. [15] released BioMistral, an open-source medical LLM fine-tuned from Mistral-7B on PubMed Central and reported competitive or superior accuracy against proprietary medical LLMs across multilingual medical QA benchmarks; its strength is reproducibility, and its weakness is that evaluation remains QA-style rather than stability-oriented. Singhal et al. [16] demonstrated with Med-PaLM and Med-PaLM 2 that large general LLMs adapted to medicine can reach 67.2% and later over 86% accuracy on MedQA (USMLE-style questions), exceeding the ~60% passing threshold, but the paper itself flags safety, bias, and uncertainty communication as unresolved issues, exactly the dimensions semantic stability addresses.

Formal grounding is supplied by UMLS literature. Bodenreider [38] described how Concept Unique Identifiers (CUIs) map many surface forms (e.g., “heart attack”, “myocardial infarction”, “MI”) to the same semantic concept, providing a well-defined equivalence relation that general-domain semantic entropy lack. Mohan and Li [28] introduced MedMentions, which annotates 4,392 PubMed abstracts with over 352,000 UMLS-linked mentions in the full corpus; the ST21pv subset, targeting 21 semantic types, contains 203,282 concept mentions, operationalizing concept-level evaluation on realistic biomedical text. Neumann et al. [29] provide ScispaCy, a biomedical NLP toolkit with UMLS entity linking that supplies the concrete preprocessing and CUI-mapping pipeline used in this study. Afshar et al. [18] showed that UMLS-aware grounding of LLM outputs improves agreement with expert judgement on diagnosis generation relative to lexical-overlap metrics such as BLEU and ROUGE, providing concrete evidence that ontology-based evaluation is methodologically stronger than string-similarity evaluation for clinical content. Dobbins [19] reported that even recent LLMs struggle with multistage biomedical concept normalisation, with F1 well below task-specific pipelines on standard normalisation benchmarks, which directly motivates measuring semantic instability at the concept rather than token level.

The distinctive contribution of this cluster is that it supplies a formal semantic equivalence relation, grounded in UMLS CUIs [38], that the uncertainty literature in Theme C lacks; MedMentions [28] and ScispaCy [29] turn that relation into a concrete pipeline. Domain-specific models [12, 15] provide the systems against which stability is measured, while the ontology-aware evaluation work of Afshar et al. [18] and the concept-normalisation results of Dobbins [19] show both that ontology-based measurement outperforms string-similarity scoring and that concept-level instability remains substantial in current LLMs. The trade-off is restricted generality, the formalism is biomedical, but the present study accepts this because it converts "semantic stability" from a heuristic quantity into a measurable one, and cross-domain validation in Theme F mitigates the loss of generality.

E. Reliability in Medicine and the Evaluation Illusion

Medical reliability work elevates the stakes of semantic instability from methodological to clinical. Ji et al. [3] surveyed hallucination across general NLG and proposed a taxonomy of intrinsic and extrinsic hallucination, while Huang et al. [4] updated the taxonomy specifically for LLMs; both conclude that hallucination has not been resolved by scale alone. Their strength is breadth of coverage, but neither offers a metric that ties hallucination to semantic-stability measurement, which is precisely the gap the present project addresses.

Three complementary studies expose the gap between benchmark and bedside. Singhal et al. [16] showed that medical LLMs pass benchmark thresholds on MedQA, as noted in Theme D, but Hager et al. [17] demonstrated in Nature Medicine that GPT-4 and Llama-2-70B-Chat dropped from strong MedQA-style accuracy to well below clinician-level performance on a realistic inpatient decision-making benchmark derived from MIMIC cases, with diagnostic accuracy, ordering accuracy, and management correctness errors on the order of 30–60% depending on task. Agrawal et al. [21] framed this pattern as the "evaluation illusion" of medical LLMs, arguing that static answer-level benchmarks systematically overstate clinical readiness, a direct intellectual parent of the motivation for this project. Zhou et al. [22] then showed that uncertainty-aware modelling for explainable disease diagnosis improves both accuracy and calibration over uncertainty-agnostic baselines, measured by classification accuracy and calibration scores on diagnostic tasks, empirically supporting the claim that better uncertainty quantification is clinically meaningful rather than cosmetic. Thirunavukarasu et al. [33] reviewed the broader landscape of medical LLMs and reiterated hallucination and real-world reliability as central risks, reinforcing the domain-level motivation of this study.

Scope boundaries are set by two further studies. Boag et al. [34] showed that clinical free-text carries predictive signal not captured by ICD-style structured codes, justifying evaluation on clinical language rather than on coded data alone. Névéol et al. [37] reviewed the multilingual limitations of clinical NLP, motivating the present study's English-only scope as a deliberate rather than incidental choice.

Read together, these studies provide the practical warrant for this project. Hallucination remains an open reliability issue in both general NLG [3] and in modern LLMs [4]. Strong medical-benchmark performance [16] does not carry over to realistic inpatient decision-making [17], which Agrawal et al. [21] frame as the evaluation illusion and Zhou et al. [22] partly address through uncertainty-aware modelling. Domain reviews of medical LLMs [33] and of clinical free-text representations [34] reinforce the clinical stakes, while the multilingual scoping of clinical NLP [37] marks the boundary within which this project operates. In medicine, therefore, semantic instability is not an abstract property but a direct source of reliability risk that the current evaluation paradigm fails to capture.

F. Cross-Domain Validation and Research Gap

To show that the proposed framework is not tied to a single dataset, cross-domain validation is built into the design. Rajpurkar et al. [27] introduced SQuAD 2.0 by adding approximately 50,000 unanswerable questions to the original 100,000 answerable ones, and reported that the best models at release dropped from ~86% F1 on SQuAD 1.1 to ~66% F1 on SQuAD 2.0, which makes the benchmark particularly informative for semantic stability: a stable system should answer consistently across paraphrased answerable items and consistently abstain on unanswerable ones. Zhu et al. [39] surveyed open-domain QA systems, benchmarks, and evaluation protocols, and position QA as a natural setting for studying semantic and epistemic uncertainty, clarifying where SQuAD 2.0 and BioASQ sit within that landscape.

Drawing the six themes together, several research gaps become explicit. First, current evaluation practice overweights benchmark accuracy despite repeated evidence that accuracy can mask instability under meaning-preserving variation, whether caused by prompt formatting [5], prompt-structure differences [6], paraphrase [8], or, in the clinical domain, benchmark overstatement [21]. Second, semantic entropy provides a principled meaning-level uncertainty metric [1], validated at scale for hallucination detection [2], but it has so far relied on general-domain NLI equivalence rather than ontology grounding such as UMLS [38]; ontology-aware evaluation improves diagnosis assessment [18] while LLMs still struggle with biomedical concept normalisation [19]. Third, biomedical LLMs [12, 15] are compared on accuracy but not systematically on semantic stability under controlled perturbation. Fourth, medical-reliability studies establish the practical problem from the benchmark–bedside mismatch of Hager et al. [17], to the evaluation-illusion critique of Agrawal et al. [21], to the broader risk review of Thirunavukarasu et al. [33], but none yet provides a structured, ontology-based measure of the evaluation illusion itself.

These gaps motivate the research direction of this study: to build a UMLS-grounded semantic entropy framework that measures the semantic stability of LLM interpretations in clinical NLP under meaning-preservation. MedMentions ST21pv supplies the primary concept-level dataset, while SQuAD 2.0 and BioASQ provide secondary validation settings. The resulting framework turns semantic stability from a qualitative concern into a measurable quantity grounded in a standard biomedical ontology and positions the project to contribute a more principled and clinically meaningful assessment of reliability in LLMs.

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